MuQ: Self-Supervised Music Representation Learning With Mel Residual Vector Quantization

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***Abstract*—Recent years have witnessed the success of foundation models pre-trained with self-supervised learning (SSL) in various music informatics understanding tasks, including music tagging, instrument classification, key detection, and more. In this paper, we propose a self-supervised music representation learning model for music understanding. Distinguished from previous studies adopting random projection or existing neural codec, the proposed model, named MuQ, is trained to predict tokens generated by Mel Residual Vector Quantization (Mel-RVQ). Our Mel-RVQ utilizes residual linear projection structure for Mel spectrum quantization to enhance the stability and efficiency of target extraction and lead to better performance. Experiments in a large variety of downstream tasks demonstrate that MuQ outperforms previous self-supervised music representation models with only 0.9 K hours of open-source pre-training data. Scaling up the data to over 160 K hours and adopting iterative training consistently improve the model performance. To further validate the strength of our model, we present MuQ-MuLan, a joint music-text embedding model based on contrastive learning, which achieves state-of-the-art performance in the zero-shot music tagging task on the MagnaTagATune dataset.**

***Index Terms*—Foundation model, music information retrieval, music understanding, self-supervised learning.**

I. Introduction

SELF-SUPERVISED learning (SSL) has been introduced into speech and audio signal processing as a technique for learning latent semantic relationships from unlabeled raw data.

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Code and checkpoints are open source in https://github.com/tencent-ailab/MuQ.

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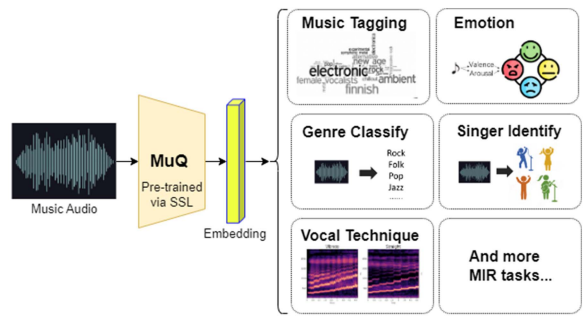

Fig. 1. Illustration of how the SSL pre-trained model works in music information retrieval (MIR) tasks.

Recently, several works [1], [2] apply SSL to music informatics understanding, and a number of pre-trained foundation models (PFMs) with generalized representation capabilities are built. Learning from unsupervised music data, these foundation models can achieve stunning performance in music understanding tasks such as genre classification, emotion prediction, and key detection [3], and further provide semantic representations for more downstream tasks like music generation [4] and music captioning [5], as illustrated in Fig. 1.

A key challenge in music understanding and representation is that music is an extremely specific modality. Unlike speech or environmental sounds, music not only focuses on semantic information, but also emphasizes acoustic information, such as melody, chords, and tonality. As a result, previous semantics-oriented SSL methods have struggled to perform well on music tasks [3], [6], [7], as they fail to simultaneously capture both semantic and acoustic information.

Recently, several studies have sought to develop a universal music representation that integrates both semantic and acoustic aspects. Pioneering works in this direction include Jukemir [8], music2vec [7] and MULE [9]. Among more recent advancements, two remarkable models are MERT [1] and MusicFM [2], both of which are mask-based self-supervised learning designs. MERT employs a BERT-style masked language modeling (MLM) proxy task to predict discrete tokens from the masked audio parts, with an Encodec [10] model as the tokenizer and uses an auxiliary Constant Q-Transform (CQT) target to enhance the modeling of acoustic information. MusicFM, on the other hand, directly utilizes a random projection quantizer derived from BEST-RQ [11], and this tokenization approach provides a general target for learning music representation, without the need for additional CQT loss to capture acoustic modeling.

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As discussed in [11] and [1], the target extractor (i.e., tokenizer) plays an important role in SSL, as models are trained to predict the tokenized pseudo-labels. BEST-RQ features a lightweight approach that allows for fast extraction of discrete targets. However, its performance is highly dependent on the initialization of the random projection layer, often requiring multiple attempts or a specific random seed to achieve optimal results. In contrast, the Encodec target [10] used in MERT produces a series of residual targets, with the multi-target strategy shown beneficial to musical SSL [1]. As a neural codec trained on audio data, Encodec produces more stable labels compared to its random counterparts. However, using Encodec as tokenization is computational heavy and consumes a lot of GPU memory when applying online extraction, which can reduce the training efficiency. Also, it needs to be coupled with additional CQT reconstruction loss to perform well in acoustic representation.

To address the initialization dependency of the random projection quantizer in BEST-RQ and the inefficiency stemming from the heavy computation cost of Encodec in MERT, we introduce a model called MuQ, which learns **Mu**sic representations from Mel **Q**uantization targets. MuQ leverages a Mel-RVQ as the tokenizer to generate targets. The proposed Mel-RVQ is pre-trained on music data and employs a linear RVQ to directly quantize audio Mel spectrograms. Compared to the random-projection quantizer in BEST-RQ, the pre-trained Mel-RVQ produces more stable targets for SSL training and eliminates the model's dependence on initialization. Additionally, compared to Encodec, the lightweight single-layer Mel-RVQ architecture offers greater extraction efficiency.

To further demonstrate the capabilities of the proposed MuQ model, we explore its application in another crucial area of music understanding: aligning music and text representations. For example, MuLan [12] employs contrastive learning to train both a music encoder and a text encoder, producing semantically consistent embeddings for both modalities and achieving coherent alignment between music and text. Recognizing the role that self-supervised learning models can play in providing effective initialization for downstream tasks, we leverage our MuQ to construct a joint music-text embedding model, named MuQ-MuLan.

Our main contributions are listed as follows:

- We introduce a novel music SSL model MuQ, which demonstrates state-of-the-art performance across a wide range of downstream music understanding tasks over previous MERT and MusicFM models.
- We propose the Mel Residual Vector Quantization (Mel-RVQ), which directly quantizes the Mel spectrum using a single linear layer RVQ, improving both training stability and efficiency.
- We further develop the MuQ-MuLan model, trained with contrastive learning on MuQ. MuQ-MuLan excels in aligning and jointly encoding music and text modalities, compared with the original MuLan model.

Our experiments demonstrate that the Mel-RVQ significantly enhances SSL performance across a variety of music downstream tasks. Notably, MuQ outperforms previous state-of-the-art (SOTA) SSL models MERT [1] and MusicFM [2]. This performance advantage is observed across different data scales, from the extensive 160K-hour dataset to the more limited 900-hour dataset, demonstrating MuQ's robustness to varying data volumes. Additionally, our results show that MuQ-MuLan achieves a ROC-AUC score of 79.3 on the MagnaTagATune zero-shot music tagging task, surpassing the previous SOTA result [12].

II. Related Work

a) Self-supervised learning for speech and audio: Self-supervised learning (SSL) [13] has been introduced to the field of speech and audio signal processing as a way to learn semantic representations from unlabeled audio data, and it is now widely used in tasks such as automatic speech recognition [14] and audio event classification [15].

SSL often relies on well-designed targets. HuBERT [16], for instance, uses K-means clustering labels to guide self-supervised learning in Masked Language Model (MLM) style, with these labels being extracted offline. BEST-RQ [11] learns directly from targets generated by a randomly initialized projection quantizer, which can be extracted online. MT4SSL [17] integrates both online and offline objectives, applying the concept of multi-target self-supervised learning, where a distinct linear prediction head is used for each type of target.

Another important topic in SSL is iterative refinement training. Most existing iterative training practices are based on clustering. For example, HuBERT applies K-means clustering on the features of an existing model to train the next iteration of models. Similarly, Seed-ASR [18] first trains an SSL model on BEST-RQ and then iteratively trains on its features using K-means clustering.

In this paper, we take an alternative approach by iteratively training directly through the quantizer itself, without the need to apply K-means clustering.

b) Music processing and understanding: Music information retrieval (MIR) contains a large set of tasks to evaluate deep learning models' understanding of music from audio signals or symbolized notation. For example, automatic music tagging task require models to do multi-label classification for tags like genre, instrument, and mood, given a music track [19]. Other MIR tasks include key detection, pitch detection, emotion analysis, etc [3].

In recent years, SSL has been introduced into music understanding for building universal representational models capable of handling various MIR tasks. Some of these efforts include CLMR [6] based on contrastive learning; MERT [1] based on MLM proxy task; and more recently, MusicFM [2]. These SSL-based models yield impressive performance across multiple MIR tasks, showing a generalized understanding of music.

c) Music-text joint embedding model: Bridging the gap between music and text modalities, music-text representation models learn joint embeddings from music-text pairwise data. Most of these models use a two-tower structure, consisting of both a music encoder and a text encoder, and are trained by contrastive learning loss.

LAION-CLAP [20] is a powerful open-source music-text encoder, with various model structures and versions trained on extensive data. MuLan [12] is also an exceptional music-text

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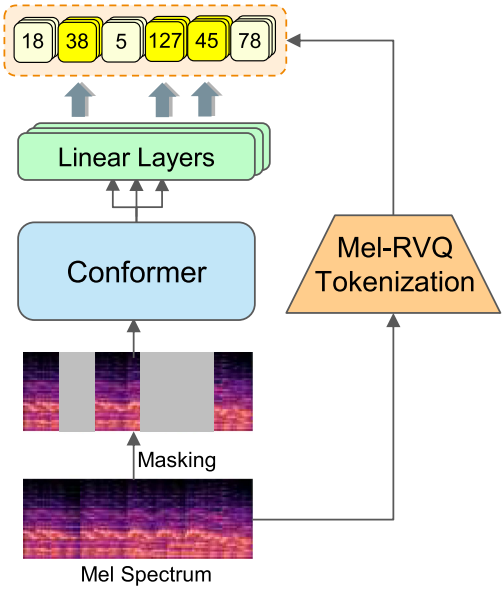

Fig. 2. An overview of our proposed MuQ's framework.

embedding model, but it is not open source for either the model or the data.

The MuQ-MuLan model introduced in this paper follows the ideas of former work but replaces the music encoder with our proposed MuQ model.

III. Method

MuQ employs a self-supervised learning approach based on masked language modeling (MLM) and tokenized targets. The overall framework of MuQ is presented in Fig. 2.

A. Self-Supervised Framework of MuQ

MuQ directly takes the Mel spectrogram of music audio sampled at 24 kHz as input. The spectrogram has 128 Mel bins, computed with a hop length of 240 and an FFT window size of 2048, and is temporally downsampled to 25 Hz. We apply random masking to the input Mel spectrum, which is partially masked as random noise with a masking probability p, and then fed into multiple layers of Conformer [21] for context learning. The Conformer output is passed through linear layers, which serve as prediction heads. Finally, we calculate the cross-entropy loss between the target and predicted labels as the optimization objective.

The target labels are tokens extracted from the Mel spectrum by the Mel-RVQ, a quantization tokenizer proposed in this paper. Since Mel-RVQ produces N tokens for each time step of the Mel spectrum, the MuQ model incorporates N distinct linear layers to predict the different target tokens. In Fig. 2, these multiple linear layers are illustrated as stacked blocks following the Conformer.

B. Mel Residual Vector Quantization (Mel-RVQ)

As shown in Fig. 3, we depict the proposed Mel Residual Vector Quantization (Mel-RVQ) compared with the random-projection quantizer in BEST-RQ and the Encodec target used by MERT.

The proposed Mel-RVQ directly takes the Mel spectrum as input and then quantizes the Mel spectrum using residual vector quantization (RVQ). The encoder of Mel-RVQ is designed as a simple single-layer linear projection, and the decoder is also a single linear layer.

Mel-RVQ needs to be trained on music data in advance before it can be used to generate quantization targets in SSL training.

1) Training of Mel-RVQ: During the training of Mel-RVQ, the final loss function can be decomposed into three terms, as shown in Fig. 3(c) :

- *Codebook loss* trains the RVQ component to improve the fit of the codebook embedding Q_τ to the dimensionality-reduced features z. Formally,

$$loss_{\text{code}} = \sum_{\substack{x\in B,\\ z=M_{\text{P}}x}} \|\text{norm}(Q_\tau) - \text{norm}(\text{sg}(z))\|^2 \quad (1)$$

  where B refers to a mini-batch of data, x denotes Mel spectrum input and sg denotes stop-gradient. M_{P} is the projection matrix and token τ is the closest to the projected vector z in the l_2-normalized embedding space.
- *Commitment loss* trains the projection (i.e. encoder) for more optimal dimension reduction and fitting to embeddings. Expressed as

$$loss_{\text{comm}} = \sum_{\substack{x\in B,\\ z=M_{\text{P}}x}} \|\text{norm}(z) - \text{norm}(\text{sg}(Q_\tau))\|^2 \quad (2)$$

- *Reconstruction loss* trains the linear decoder (denoted as M_{D}) and the codebook to restore the original feature x. The formula is

$$loss_{\text{recon}} = \sum_{x\in B} \|M_{\text{D}}Q_\tau - x\|^2 \quad (3)$$

The final loss used to train the Mel-RVQ is the weighted sum of the above three losses:

$$loss = \alpha \cdot loss_{\text{code}} + \beta \cdot loss_{\text{comm}} + loss_{\text{recon}} \quad (4)$$

where α and β are weighting factors, set as $\alpha = 1, \beta = 0.25$.

It is emphasized that both M_{D} and M_{P} used in Eq.(4) are simple linear layers, which distinguishes them from those in neural codecs like Encodec [10].

2) Residual Modeling of Mel-RVQ: Applying the residual modeling method to Mel-RVQ means that it yields multiple tokens for each time step of the audio. Assume there are N codebooks in total, denoted as $\{Q_i^{(n)}\}_{i=1}^K$ for $n \in 1 \ldots N$. In this way, we have

$$\begin{aligned} z^{(n)} &= M_{\text{P}}^{(n)} r^{(n-1)} \\ \tau^{(n)} &= \arg\min_i \|\text{norm}(z^{(n)}) - \text{norm}(Q_i^{(n)})\| \\ r^{(n)} &= r^{(n-1)} - M_{\text{D}}^{(n)} Q_{\tau^{(n)}}^{(n)} \end{aligned} \quad (5)$$

where $n \in 1 \ldots N$; $r^{(n)}$ is the residual signal passed to next step in quantizer and $r^{(1)} = x$. $M_{\text{P}}^{(n)}$ and $M_{\text{D}}^{(n)}$ denote the residual-projection matrices for the encoder and decoder, which contain

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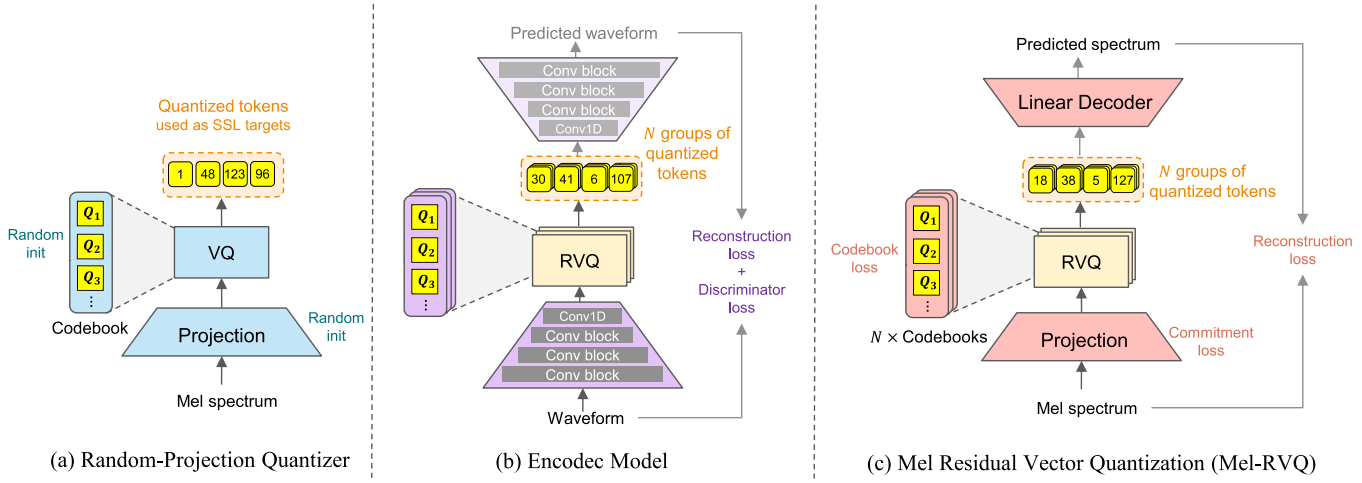

Fig. 3. Comparison between (a) random-projection quantizer, (b) Encodec, and (c) proposed Mel Residual Vector Quantization (Mel-RVQ). Mel-RVQ draws on the lightweight structure of the random-projection quantizer, i.e., it uses only a single linear projection as the encoder, with the Mel spectrum as input. Also, Mel-RVQ borrows the proven effective residual structure from Encodec, but Mel-RVQ further benefits from a much simpler structure than the multiple convolutional layers in Encodec.

multiple steps of projection. The superscript (n) denotes the components or features corresponding to the quantizer at step n.

3) Iterative Refinement With RVQ: Iterative refinement is introduced in HuBERT [16] as a method to improve the performance of SSL models, where clustering techniques like K-means are employed to produce new labels for the next iteration of training.

We suggest that residual vector quantization (RVQ) itself can serve as an alternative to the clustering in iterative enhancement. That is, we directly train a Mel-RVQ$_{\text{iter}}$ on the latent representations of an already trained MuQ model, to re-drive the training of MuQ at next iteration.

Specifically, the iterative training on MuQ has two stages:

- In the initial stage, a Mel-RVQ is trained on the Mel spectrum feature, and then the first version of MuQ is trained on the tokens produced by Mel-RVQ.
- In the iterative stage, the first version of MuQ is used to extract representation from the audio, and the l th layer latent is used to train Mel-RVQ$_{\text{iter}}$, which will be used in training the second version of MuQ.

C. Music-Text Contrastive Learning

To verify the effectiveness of MuQ in more sophisticated downstream tasks, we trained a music-text joint embedding model, MuQ-MuLan. Similar to MuLan [12], MuQ-MuLan employs a two-tower multimodal architecture and is trained on a large amount of (music, text) paired data.

MuQ-MuLan consists of a music encoder and a text encoder. During training, the music and the corresponding text are fed to the encoders separately, where the text is a description of the music track, as shown in Fig. 4.

For music modality inputs, a MuQ model with pre-trained parameters is first used to encode the audio into a latent representation. This representation is then average-pooled over the temporal dimension. Finally, it passes through a linear projection layer to produce a music embedding e_{m} with dimension d.

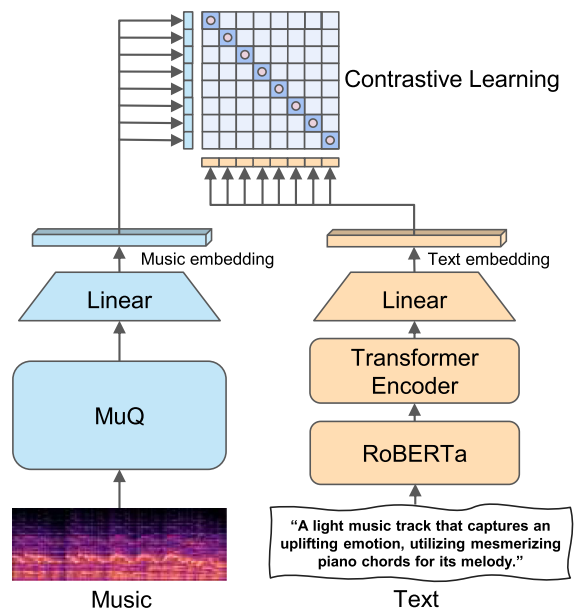

Fig. 4. Overall architecture of MuQ-MuLan, a music-text joint embedding model trained with contrastive learning.

For text modality inputs, they are encoded by a pre-trained RoBERTa [22] model. As described in a previous study [23], the text encoder is critical to the performance of the two-tower model, so we append a few additional layers of Transformer encoder after RoBERTa. Likewise, the textual latent representations are average-pooled and passed through a linear projection layer to get text embedding e_{t} with the same dimension d.

We use decoupled contrastive learning (DCL) loss [24]. Formally, given the i-th (music, text) pair in a mini-batch, for the embeddings $(e_{\text{m}}^{(i)}, e_{\text{t}}^{(i)})$, we have

$$L_{\text{DCL}} = -\log \sum_{(i,j)} \frac{\text{sim}(e_{\text{m}}^{(i)}, e_{\text{t}}^{(j)})/\sigma}{\sum_{i \neq j} \text{sim}(e_{\text{m}}^{(i)}, e_{\text{t}}^{(j)})/\sigma} \tag{6}$$

where sim denotes dot-product similarity, and σ is temperature coefficient.

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IV. Experiments Setup

A. Model Settings

1) Muq: We stack 12 layers of Conformer in MuQ, with 310 M parameters in total. The number of codebooks for Mel-RVQ is set to $N = 8$ and the codebook size is $K = 1024$. The input audio is fixed at 30 seconds with a sampling rate of 24 KHz. The hop length of the Mel spectrogram is 240, with an FFT window size of 2048 and 128 bins. Before being fed into the Conformer, the Mel spectrogram input is downsampled to 25 Hz by a pooling layer. The code is implemented with the Fairseq toolkit [25].

2) Muq-Mulan: We use the pre-trained MuQ as the initialization for the music encoder. A pre-trained multilingual RoBERTa model `xlm-roberta-base` is used for the text encoder followed by 8 layers of Transformer, with the embedding dimension $d = 512$. MuQ-MuLan has 630 M parameters in total, and all parameters stay unfrozen during the contrastive training.

B. Pre-Training

1) Dataset: We initially conduct pre-training on a smaller dataset, Music4all [26], containing 0.9 K hours of open access music data. Subsequently, we scale up to a large-scale in-house dataset with 160 K hours of music. We also train our model on the Million Song Dataset (MSD) [27], a large public music dataset containing 1 million songs, to enable reproducible comparisons with baselines.

2) Training Details: Mel-RVQ is trained on the Music4all dataset using 1 GPU. Since the Mel-RVQ is very small (less than 0.3 M), the training of Mel-RVQ is extremely fast and it takes less than 1 h to complete.

All pre-training experiments are running with 32 A100-40 G GPUs. Batch size is 192, and masking probability $p = 0.6$. For the pre-training on the Music4all dataset, we train for only 75K steps. For the experiments on the 160K hours of in-house data, we initially train MuQ for 200K steps. Subsequently, the $l = 10$th layer latent is used to train $\text{Mel-RVQ}_{\text{iter}}$, followed by 150K steps of iterative training from scratch to obtain MuQ_{iter}. The entire initial & iterative pre-training takes about 2 weeks.

C. Contrastive Training

1) Dataset: The training data for contrast learning comes from 130 K hours of music, each with its corresponding text description. The dataset comprises three components: the MTG-Jamendo Dataset [28], the Million Song Dataset [27] with LP-MusicCaps-MSD annotations [29], and a web-collected music collection. The text descriptions are limited to non-personal information such as genre, mood, instrument, tempo, keywords of the music. A detailed description of this dataset can be found in the Section V(c).

2) Training Details: MuQ-Mulan is trained on 32 V100-32 G GPUs with a batch size of 768. Following the practice of MuLan [12], we switch to taking 10 seconds as the length of the input audio. This significantly reduces memory consumption and enlarges batch size, which is critical for contrastive learning.

D. Downstream Tasks

We evaluate MuQ on the MARBLE benchmark [3] and conduct experiments on a total of 9 downstream tasks. Most of the downstream tasks are based on the MARBLE-benchmark [3] implementation. Note that all datasets appearing in the downstream tasks are **not** included in the pre-training data of MuQ. An overview of the downstream tasks are provided in Table II, including their dataset, prediction type, and number of categories (or dimensions of the regression target).

- *Genre classification* aims to determine the most suitable genre for each song in dataset. We use the GTZAN dataset [30] for this task. The GTZAN dataset has a collection of 10 genres with 100 audio files each, all having a length of 30 seconds. We set windows of 10-seconds as inputs to get the features respectively, then combine the results to get the features used for the downstream task. We utilize the standard "fail-filtered" split [31] for training and evaluating, then we provide the accuracy scores on testing set to show the final result.
- *Key detection* estimates the tonal scale and dominant pitch level of each given song. Specifically, the labels of the task contains major and minor scales for all pitch classes, that is, 24 classes in total. We use the Giantsteps dataset [32], which has a collection of annotations for 604 2-min audio previews of songs, as the test set, and a commonly-used subset of the Giantsteps-MTG-keys dataset [33], for training and validation. We use the refined accuracy with reasonable error tolerance score [34] to evaluate the performance of models.
- *Emotional analysis* The Music Emotion Recognition aims to get the Emotion score of songs. We use the Emomusic dataset [35], which contains 744 music clips of 45 seconds. Each clip is labeled with two scores: valence score indicates positive and negative emotional responses, and arousal score indicates emotional intensity. The dataset split is same as [8]. We divide the whole signal into 5 s windows as the inputs of SSL models to get the features, then combine the features as the inputs of a simple regression models. We calculate the determination coefficient (R2) between the model regression results and human annotations of arousal (R2A) and valence (R2V) [35] as metrics for the test results.
- *Singer identification* is to recognize the corresponding singer based on the given singing audio. We use the Vocalset dataset [36], which contains 20 different singers (11 male and 9 female) who perform 17 different singing techniques in various contexts. We randomly split the dataset into traning,validation and testing sets based on a ratio of 12:8:5, all of which contain the full 20 singers. The evaluation metric is accuracy.
- *Vocal technique detection* is to identify the techiques that the singer used in a given audio segement. We use the VocalSet dataset [36] for this task. As the audio clips are divided into 3 seconds, the task only requires a judgement on the type of technique and not on the start and end of the technique. We used the same 10 different singing techniques as in [37] as a subset and used the same 15

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singers as the training and validation sets and 5 singers as the test set. For evaluation purposes, we employ accuracy as the metric in this classification task.

- *Music tagging* assigns a set of fixed tags for a given song. The tags contain genre, instrumentation, mood and so on. We use the MagnaTagATune (MTT) [38] dataset. MTT dataset contains 25,863 music clips. Each clip is a 29-seconds-long excerpt belonging to one of the 5223 songs. Each audio clip is labeled by human with a vector of binary annotations of 188 tags like "singer", "no singer", "violin", "drums", "classical", "jazz". The top 50 most popular tags are typically used for evaluation to ensure that there is enough training data for each tag. The metrics are the macro-average of ROC-AUCs, a multiclass AUC as the average of the area under multiple binary ROC curves, and the average precision (AP) / PR-AUC, the weighted mean of precisions at each threshold, among all top-50 tags.
- *Music structure analysis* splits a music signal into non-overlapping sections and predicts functional labels like "intro", "verse", "chorus". We use the Harmonix Set [39] for this frame-wise classification task. First, we construct probability curves from data annotations. We standardize labels following [40]. We use a simple LSTM model to predict frame-wise probabilities and use frame-wise accuracy for functional labels as the metric.
- *Instrument classification* Instrument classification identify which instruments are used in a given music segment. We use the NSynth dataset [41]. The NSynth Dataset is an audio dataset containing 300 k musical notes, each with a unique pitch, timbre, and envelope. Each segment may contain multiple instruments. We evaluate the model using ROC and AP scores.
- *Pitch classification* Pitch classification aims to detect the certain pitch of a given short audio clip in 128 pitch categories. We use the NSynth dataset [41] for this task. We use accuracy score to evaluate the performance.

In evaluating the pre-trained MuQ and other SSL models, we extract features from each layer and use a linear probe to perform downstream tasks. For the MuQ-MuLan experiments, we use a zero-shot approach for music tagging. This involves directly calculating the similarity between music and text embeddings to determine the prediction probability for each tag.

E. Baseline Models

In order to evaluate the effectiveness of our proposed method, we compare it to representative audio representation models and describe each model here.

- *MERT [1]* is an SSL model pre-trained on large-scale music datasets using masked prediction tasks. MERT employs a CNN frontend to process waveform signals and extract 75 Hz features, which are then stacked with multiple Transformer layers. Its objectives include minimizing the cross-entropy loss of the codes extracted by Encodec and the reconstruction loss of the CQT spectrogram. It uses the codes extracted by EnCodec as a substitute for the K-Means targets in the HuBERT [16], and leverages the CQT loss to capture acoustic information.
- *MusicFM [2]* is a music SSL model based on mask prediction. It replaces the Transformer with the Conformer [21] as the backbone and operates on 25 Hz feature. MusicFM employs a random projection quantizer as its target extractor, similiar to BEST-RQ [11]. Compared to MERT, MusicFM has a simpler training objective, while achieving strong semantic and acoustic capabilities.
- *LAION-CLAP [20]* uses a CLIP-like contrastive learning paradigm to build large-scale language-audio pre-training models. It uses HTSAT [42] as the text encoder and RoBERTa [22] as the audio encoder, and is trained on a collection of 2.5 M audio-caption pairs from LAION-Audio-630 K dataset.
- *Microsoft-CLAP 2023 [43]* is a large contrastive pretrained model that is trained on 4.6 M audio-text pairs, using HTSAT as the audio encoder and GPT2 as the text encoder.
- *MuLan [12]* is a contrastive learning model tailored for music and text embeddings. It applys AST [44] as music encoder and BERT [45] as the text encoder, trained on 44 million music tracks (370 K hours).

The proposed model in this paper differs from previous models in the following aspects:

For the SSL model, while MuQ adopts the same Conformer backbone as MusicFM, it employs a different training objective from both MERT and MusicFM. Specifically, MuQ is trained using the proposed Mel-RVQ, which is a small target extractor that does residual quantization directly on the Mel spectrum.

For the contrastive learning model, unlike previous work, MuQ-MuLan uses MuQ as the music encoder and RoBERTa as the text encoder, aiming to validate the generalization capability of MuQ in the context of contrastive learning.

F. Performance of Pre-Trained MuQ

We compare MuQ with two popular models, MERT and MusicFM. For MERT, we use `MERT-v1-330 M` version [1], the most superior of its releases on HuggingFace. We also use its released `MERT-v0-public` version for comparison, which is trained on the Music4all [26]. For MusicFM, we use the checkpoint pre-trained on the MSD dataset [27], which is publicly available by [2]. We also trained a MusicFM model in the 160K-hour in-house dataset to assess the performance of the MusicFM baseline in large-scale dataset scenarios. For all SSL models, we evaluate each individual layer on the downstream task and report the result of the best-performing layer. All MuQ and MusicFM models are of 310 M parameters.

Table I present the results for MuQ and SSL beselines. When trained on 160 K hours of large-scale data, MuQ achieves superior performance over MusicFM in 8 out of 9 tasks, and demonstrates higher average scores than both `MERT-v1-330 M` (77.0 vs. 74.5) and MusicFM (77.0 vs. 76.0). We visualize the results of MuQ compared to other open source models in Fig. 5.

MuQ also excels even when trained in small-scale datasets. When trained on just 0.9 K hours of Music4all data, MuQ

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TABLE I
EVALUATION OF PROPOSED MUQ MODEL IN MARBLE BENCHMARK

Pretrain Dataset		Dataset	GTZAN	GS	EMO		VocalSet		HX	MTT		Nsynth		Avg.
		Task	Genre	Key	Emotion		Singer	Tech	Structure	Tagging		Instrument	Pitch	
		Metrics	Acc↑	$Acc^{Refined}$↑	$R2^V$↑	$R2^A$↑	Acc↑	Acc↑	Acc↑	ROC↑	AP↑	Acc↑	Acc↑	
Open-Source Dataset	Music4all Dataset (0.9K hours)	MERT-v0-public	73.2	63.4	57.3	74.7	78.4	73.0	75.6	90.9	38.8	70.4	92.3	71.6
		MusicFM	77.9	62.8	56.4	71.8	95.5	76.3	77.0	90.8	39.0	74.6	90.7	73.9
		MuQ	84.0	63.7	60.0	76.0	95.2	80.8	76.4	91.1	39.0	76.9	91.2	75.8
	MSD Dataset	MusicFM	83.8	63.9	60.3	76.3	92.0	78.4	76.5	91.3	40.0	76.2	91.1	75.4
		MuQ♯	85.5	64.4	61.4	**78.5**	94.3	79.0	**77.3**	91.3	39.8	**80.1**	91.4	76.6
Private Dataset	160K hours From Internet	MERT-v1-330M	79.3*	**65.6***	61.2*	74.7*	87.1*	76.9*	75.7	91.3*	40.2*	72.6*	**94.4***	74.5
	160K hours In-house	MusicFM	83.4	63.5	58.5	76.0	95.6	81.1	76.5	91.2	39.9	79.1	91.6	76.0
		MuQ	85.5	63.5	60.9	75.4	95.8	**82.4**	76.5	**91.4**	**40.3**	79.2	92.3	76.7
		MuQ♯	**85.6**	65.0	**62.8**	76.1	**96.2**	81.6	77.2	**91.4**	40.1	79.7	91.3	**77.0**

MuQ♯ denotes MuQ model using iterative refinement training. * indicates that the number is brought from the original paper.

TABLE II
DESCRIPTIONS OF DOWNSTREAM TASKS USED IN OUR EXPERIMENTS

Dataset	Task	Prediction Type	Metrics	# Categories
GTZAN	Genre Classification	Multi-class	Acc↑	10
Giantsteps(GS)	Key Detection	Multi-class	$Acc^{Refined}$↑	24
Emomusic(EMO)	Emotion Analysis	Regression	$R2^V$↑ / $R2^A$↑	2-dim
VocalSet	Singer Identification	Multi-class	Acc↑	17
VocalSet	Vocal Technique Detection	Multi-class	Acc↑	10
Harmonix(HX)	Music Structure Classification	Multi-class	Acc↑	9
MagnaTagATune(MTT)	Music Tagging	Multi-label	ROC↑ / AP↑	50
Nsynth	Instrument Classification	Multi-class	Acc↑	11
Nsynth	Pitch Classification	Multi-class	Acc↑	128

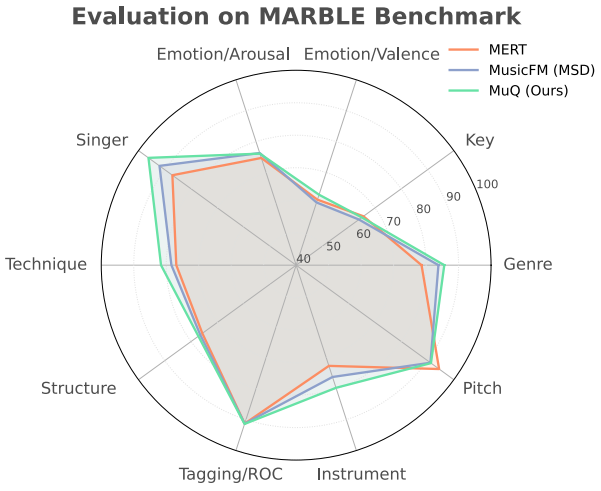

Fig. 5. Visualization of the results of MuQ compared to other open source models (`MERT-330 M-public` and MusicFM trained on MSD) on the MARBLE benchmark.

attains an average score of 75.8, outperforming comparable models. Remarkably, the MuQ model trained in Music4all achieves results comparable to those trained in large-scale dataset such as `MERT-v1-330 M` and MusicFM. This shows that MuQ achieves excellent data efficiency while demonstrating robust performance across diverse dataset sizes.

MuQ excels on tasks such as GTZAN (Genre), VocalSet (Singer), and HX (Structure). We believe it is because these tasks benefit from fine-grained token representations and global temporal consistency, which are supported by MuQ's residual modeling tokenizer. Table I also shows that iterative refinement improves greatly on tasks like HX (Structure). This is because genre and singer recognition require capturing subtle timbral cues, while structure prediction depends on stable, temporally-aware representations.

We also note that the number of output categories in downstream tasks (as shown in Table II) does not always correlate negatively with the performance. For example, the Pitch Classification task involves 128 categories, yet our model achieves a high accuracy (91.3%), while the GTZAN genre classification (10 categories) has a lower score (85.6%). This suggests that other factors such as the label quality or acoustic distinctiveness might play a larger role than the category.

G. Ablation Study on Mel-RVQ

We carry out an ablation study for Mel-RVQ in Table III to demonstrate the effect of its key components. In the ablation experiments, all models are pre-trained 75 K steps on the Music4all with 310 M parameters, and tested on 5 tasks.

The VQ type in Table III indicates whether the VQ is randomly initialized or trained on music data, and Residual indicates whether residual multiple codebooks are used. As shown, it's crucial to train the VQ rather than leaving it random, as this boosts the performance of MusicFM and MuQ by 1.6 and 2.8, respectively. Also, the residual structure of Mel-RVQ is beneficial and allows MuQ to gain a significant advantage over MusicFM.

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TABLE III
ABLATION STUDY OF DIFFERENT COMPONENTS IN MEL-RVQ

Model codebase	Components		GTZAN	GS	EMO		VocalSet		Average
	VQ type	Residual	Genre Acc↑	Key $Acc^{Refined}$↑	Emotion $R2^V$↑	$R2^A$↑	Singer Acc↑	Tech Acc↑	
MusicFM	Random	✗	77.9	62.8	56.4	71.8	95.5	76.3	73.5
	Trained	✗	80.0	**65.8**	56.3	74.6	95.3	78.5	75.1
MuQ	Random	✓	77.6	62.1	56.5	74.1	**95.7**	76.5	73.8
	Trained	✓	**84.0**	63.7	**60.0**	**76.0**	95.2	**80.8**	**76.6**

All models are pre-trained on the Music4all dataset.

TABLE IV
ABLATION STUDY OF TARGET STRATEGY

Tokenizer	Num. of codebooks	GTZAN	GS	EMO		VocalSet		Average
		Genre Acc↑	Key $Acc^{Refined}$↑	Emotion $R2^V$↑	$R2^A$↑	Singer Acc↑	Tech Acc↑	
Mel-RVQ	1	82.8	45.9	55.3	74.6	93.7	75.4	71.3
Mel-RVQ	4	78.3	65.1	59.0	73.7	95.2	79.5	75.1
Mel-RVQ	8	**83.4**	**65.8**	**59.7**	**76.2**	**95.7**	**80.1**	**76.8**
Mel-RVQ (deeper)	8	81.7	64.4	55.9	72.3	95.5	80.4	75.0
Encodec	8	82.4	65.3	57.1	74.1	95.2	77.9	75.3

All models are pre-trained on Music4all dataset.

TABLE V
EXPERIMENT ON THE MAGNATAGATUNE ZERO-SHOT MUSIC TAGGING TASK

Model	Data (hours)	Metrics	
		ROC-AUC↑	PR-AUC↑
LAION-CLAP	4K	73.9	25.0
Microsoft-CLAP 2023	-	75.9	28.9
MuLan	370K	78.2*	-
MERT-MuLan	130K	75.5	24.6
MusicFM-MuLan	130K	76.2	27.1
MuQ-MuLan	130K	**79.3**	**29.3**

Since MuLan is a closed source, the results of MuLan are taken from the paper [12].

H. Ablation Study on Music SSL Target

We ablate the number of target tokens and tokenization SSL training, this involves some hyperparameters of Mel-RVQ, as well as a comparison with Encodec target.

As seen in Table IV, performance gradually improves as the number of codebooks increases. We believe this is because more codebooks allow the model to capture fine-grained audio features more effectively. This also implies the advantage of multi-target learning for SSL pre-training.

We also explore the depth of the Mel-R VQ encoder and decoder by scaling up the linear projection to a 3-layer fully connected network, referred to as Mel-RVQ (deeper) in Table IV. However, the results suggest that increasing the depth of the Mel-RVQ network actually degrades performance. This indicates that a single layer of linear projection is sufficient for Mel-RVQ to achieve optimal performance.

Finally, we compare Mel-RVQ with the Encodec target. It can be seen in Table IV that Mel-RVQ, despite being a very lightweight structure, outperforms the Encodec tokenization target. We posit that this is because the structure and parameter count of Encodec's encoder and decoder are overly complex, resulting in tokens that represent a highly abstracted version of the original audio. This abstraction may make them difficult for SSL models to learn. In contrast, Mel-RVQ, with its simple architecture, generates tokens that more closely resemble straightforward clusters of the original Mel spectrogram, thus facilitating easier learning.

This reveals that Mel-RVQ is able to strike a good balance between the complexity of the target and the acoustic modeling of the music. Despite its simple structure, Mel-RVQ demonstrates the ability to extract meaningful acoustic and semantic patterns from music, even though its performance on certain tasks still lags behind domain-specific supervised models.

I. Zero-Shot Performance of MuQ-MuLan

In Table V, we compare MuQ-MuLan with LAION-CLAP [20], Microsoft-CLAP 2023 [43] and MuLan [12]. The details of these models above are described in Section IV-E. We use the open-source `larger_clap_music_and_speech` version for LAION-CLAP, which is the best-performing checkpoint among its releases in our experiments. We also introduce two variants that employ MERT or MusicFM as the music encoder and refer to them as MERT-MuLan and MusicFM-MuLan.

Following previous study [23], we choose the proven superior MusCALL prompt: "A <label> track" for all models to test in our experiments.

MuQ-MuLan outperforms previous models in both ROC-AUC and PR-AUC and reaches the SOTA in the zero-shot setting. Among them, the representatives of existing open source models, LAION-CLAP and Microsoft CLAP 2023, achieve ROC-AUC of 73.9 and 75.9, respectively, while the proposed MuQ-MuLan can achieve a superior performance of 79.3 ROC-AUC. In addition, MuQ-MuLan still has a slight advantage over MuLan despite the gap in data amounts. Given the unavailability of MuLan's training details and data, we assume that this advantage comes from multiple factors such as the training architecture and data of our MuQ-MuLan, as well as a better initialization provided by MuQ.

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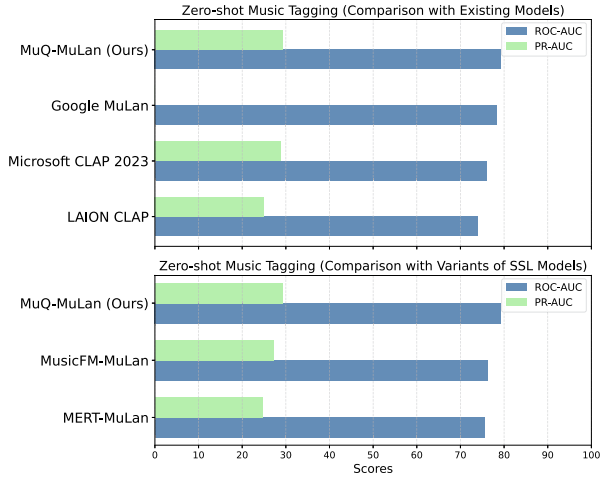

Fig. 6. Visualization on zero-shot music tagging task in MagnaTagATune dataset.

TABLE VI
ZERO-SHOT EVALUATION RESULTS OF GENRE, INSTRUMENT, AND MOOD CLASSIFICATION ON THE MTG DATASET, AND GENRE CLASSIFICATION ON GTZAN DATASET

Dataset	**MTG**						**GTZAN**	
Task	**Genre**		**Instrument**		**Mood**		**Genre**	
# Categories	95		41		59		10	
Metrics	ROC↑	PR↑	ROC↑	PR↑	ROC↑	PR↑	Acc^{top1}↑	Acc^{top3}↑
LAION-CLAP	81.4	27.4	**78.4**	39.9	68.7	19.3	63.8	84.5
MS-CLAP 2023	69.6	15.4	77.0	38.0	62.0	15.6	53.4	79.0
MuQ-MuLan	**85.8**	**32.4**	78.0	**40.9**	**77.1**	**22.6**	**78.6**	**94.8**

Although both use the same contrastive learning setup, MuQ-MuLan outperforms MERT-MuLan. Furthermore, even though MuQ and MusicFM have identical model structures, MuQ-MuLan still performs better. This suggests that MuQ provides a better initialization for contrastive training. It also shows how better SSL models can facilitate better contrastive learning models, allowing learning to a shared space that aligns music and text.

We also visualize the performance of MuQ-MuLan in Fig. 6, and plot the comparision with LAION-CLAP [20], Microsoft-CLAP 2023 [43], MuLan [12] and two variants of SSL models.

J. Extended Evaluation of MuQ-MuLan

To further assess the generalization ability of MuQ-MuLan, we expand our evaluation to include downstream classification tasks from the MTG dataset (genre, instrument, and mood) and the GTZAN dataset (genre). We use the standard test split and use the same evaluation procedure and text prompt as in Section IV-I. Results are shown in Table VI.

While MTG-Jamendo is partially included in the MuQ-MuLan pretraining data (approximately 5% of the corpus), we ensure that the evaluation test split has no overlap with the training split. Although this setup is not strictly zero-shot, it still offers a meaningful measure of generalization. In contrast, GTZAN is entirely unseen during training and serves as a true zero-shot benchmark.

As shown in Table VI, MuQ-MuLan consistently outperforms LAION-CLAP and Microsoft-CLAP 2023 across nearly all tasks. It achieves the best ROC-AUC and PR-AUC on MTG-Genre (85.8 / 32.4), as well as the highest top-1 and top-3 accuracy on GTZAN (78.6 / 94.8). Performance gains on MTG-Instrument and MTG-Mood further demonstrate its versatility across music tagging tasks.

These results support MuQ-MuLan as a general-purpose foundation model for zero-shot or near zero-shot music understanding.

K. Stage-Wise Visualization of Mel-RVQ

To better understand the internal behavior of Mel-RVQ, we visualize the decoder outputs at different quantization stages using a representative music audio clip. Specifically, we extract and display the residual signal decoded at each quantization stage, as well as the cumulative reconstructed Mel spectrograms after applying the first 1/8, 3/8, 5/8, 7/8, and 8/8 codebooks.

As shown in Fig. 7, we observe a clear coarse-to-fine refinement process across stages:

- Early stages (e.g., 1/8) primarily reconstruct the overall energy contour and broad spectral shape. The decoded residuals at this stage are relatively uniform, matching the general sketch of the target Mel spectrogram.
- Middle stages (e.g., 3/8 to 7/8) begin to add structured harmonic content and texture, with residual energy concentrated in specific frequency bands (highlighted with dashed boxes in Fig. 7), particularly in low and mid frequencies. These patterns reflect more precise modeling of timbral and tonal characteristics. This corresponds to the emergence of more clearly defined harmonics in the reconstructed spectrograms in Fig. 7(g)–(h).
- Later stage (e.g., 8/8) contributes widespread refinements, evenly distributed across frequency bins, and helps correct residual errors while enhancing local spectral details. In the reconstructed spectrogram (Fig. 7(i)), it helps improve overall fidelity and texture consistency, particularly by complementing high-frequency details.

While no explicit frequency decomposition is enforced in Mel-RVQ, this behavior suggests that its residual quantization mechanism implicitly organizes spectral information from coarse global structure to fine-grained localized detail. This analysis provides an interpretable view of how Mel-RVQ progressively constructs rich audio representations through stage-wise decoding.

L. Layer-Wise Analysis

In Fig. 8, we visualize the results of each layer's features in various downstream tasks during one run of the MuQ model. The analysis can be summarized as follows:

- *Semantic tasks*, such as genre classification and structure analysis, generally achieve the best results at the higher layers. They focus more on semantic information such as the direction of the music and the overall style.

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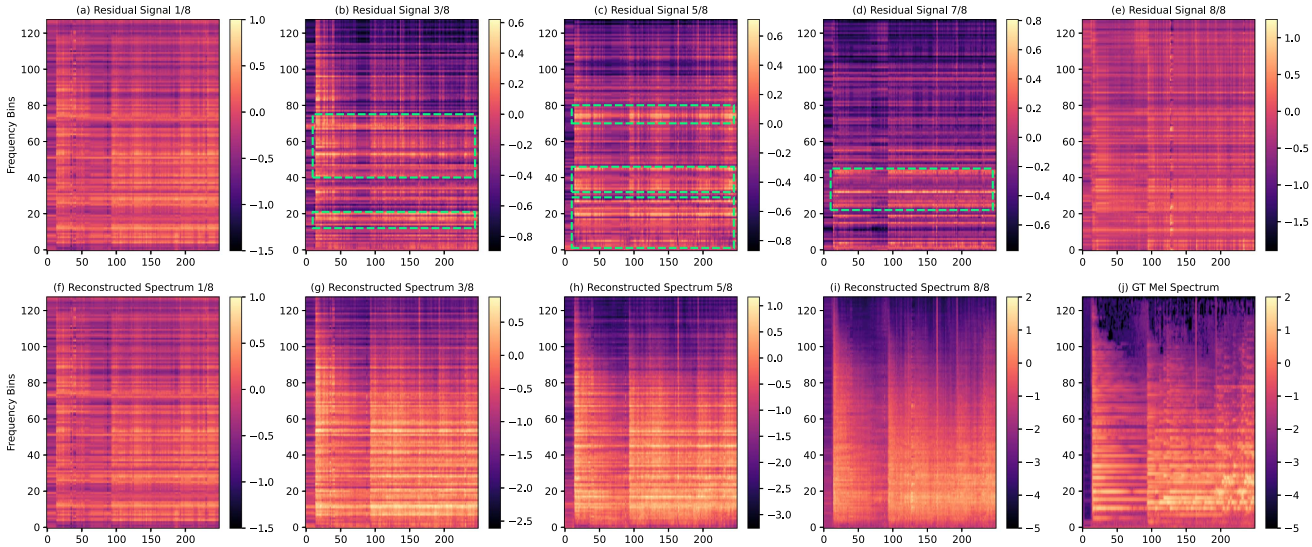

Fig. 7. Visualization of residual signals and reconstructed Mel spectrograms at different quantization stages of Mel-RVQ (1/8, 3/8, 5/8, 7/8, and 8/8). (a–e): Residual signals added at each stage; concentrated regions are highlighted with dashed boxes. (f–i): Corresponding cumulative reconstructed spectrograms after applying up to that stage. (j): Ground-truth Mel spectrogram.

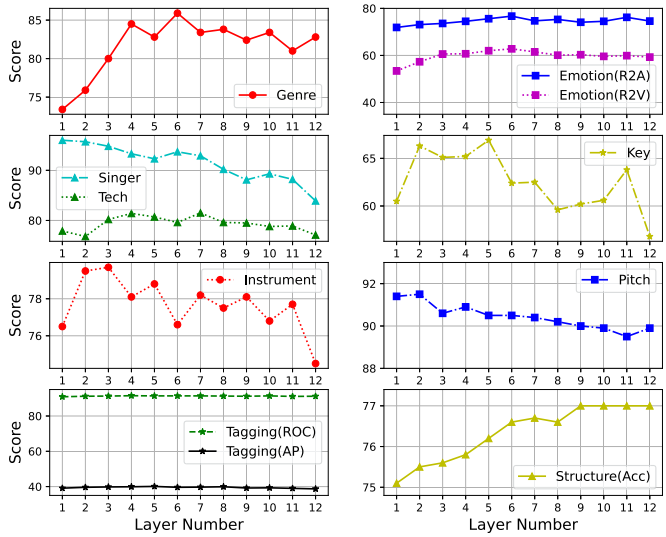

Fig. 8. Layer-wise results of MuQ in MARBLE benchmark.

- *Acoustic tasks*, such as singer identification, instrument classification, pitch classification, and key detection, perform best at the lower layers. This implies that they are more closely associated with low-level characteristics such as the original audio features.
- *Comprehensive tasks*, such as music tagging, emotional analysis, and vocal tech classification, have a relatively even distribution across all layers. This indicates that they are dependent on various aspects, not limited to mere semantic or acoustic features.

V. Discussion

a) Additional explanation of Mel-RVQ: Although in this paper, we only discuss the case of Mel-RVQ with up to 8 codebooks, this does not mean that more codebooks are meaningless. Exploring Mel-RVQ with more codebooks may help understand the reasons why Mel-RVQ works and the limits of this advantage. The selection of layer 10 in iterative training is based on empirical intuition rather than rigorous experimental comparison. Due to limited computational resources, we did not explore other layers. We believe that iterating over other layers could potentially lead to better results, and this is a worthwhile topic of future research.

TABLE VII
Comparison of Parameters Number and FLOPS for Different Quantizers

Tokenizer	Params	FLOPS
Encodec	22.196 M	44.730 G
Random Projection Quantizer	0.023 M	0.006 G
Mel-RVQ	0.266 M	0.098 G

b) Computational efficiency of Mel-RVQ: The computational efficiency of Mel-RVQ can be demonstrated by comparing parameters and FLOPS. As shown in Table VII, we compare Mel-RVQ with Encodec and the Random Projection Quantizer. Mel-RVQ is close to the Random Projection Quantizer in terms of the number of parameters and the amount of operations, and is much smaller than Encodec. Mel-RVQ outperforms Encodec in both parameters and computation, with efficiency comparable to random projection, while delivering the best SSL performance.

c) Training dataset and reproducibility of MuQ-MuLan: Our contrastive learning model, MuQ-MuLan, is trained on a curated dataset composed of three main sources. First, we incorporate the MTG-Jamendo dataset [28], which is licensed under Creative Commons and commonly used for music tagging tasks. Second, we include the Million Song Dataset (MSD) [27], with the LP-MusicCaps-MSD [29] annotations to provide text description. Lastly, approximately 77% of our dataset consists of a

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TABLE VIII
STATISTICS OF THE DATASET FOR CONTRASTIVE LEARNING TRAINING

Category	Value
Total duration	∼130K hours
Total tracks	∼3.3 M
Data source composition	
MTG-Jamendo Dataset	5%
Million Song Dataset (MSD)	18%
Web-collected music	77%
Text description coverage	
Genre tags	112 types
Instrument tags	77 types
Free-form tags	∼16,000 types
Language	English, Chinese
Audio characteristics	
Average duration	141.6 seconds
Sampling rate	44,100 kHz

web-collected music corpus gathered from publicly available sources. This portion is used strictly for non-commercial academic research. To promote transparency and reproducibility, we provide a complete statics of our data in Table VIII. For readers interested in replicating our work, we also recommend two open-source alternatives: LAION-Audio-630K[1] [20] and M4-RAG[2] [46], which contain large-scale music-text pairs suitable for contrastive learning. In the data processing pipeline, we employ two distinct approaches to extract free-form text from tags: comma-delimited concatenation of tags to form coherent sentences, and structured template integration using predefined prompts. To ensure bilingual compatibility, all English tags are subsequently translated into Chinese using a large language model. This dual-language processing yield parallel datasets in both English and Chinese, facilitating cross-linguistic application.

d) Limitation and future work: Mel-RVQ is required to be pre-trained on music data before being used for SSL targeting. While we emphasize that the training process for Mel-RVQ is very fast (less than 1 h), this is still not a completely out-of-the-box approach like BEST-RQ. In future work, we will explore different model sizes (e.g., 90 M *base* size) for the requirements of different physical environments and computational conditions. We will also be working on using MuQ for generative tasks such as music generation.

VI. CONCLUSION

In this study, we introduce MuQ, a novel self-supervised learning model for music representation and understanding. Building on this, we also present MuQ-MuLan, a joint music-text embedding model. MuQ uses Mel-RVQ as a quantization target, applying a linear projection RVQ structure to the Mel spectrum, which provides a simple yet powerful SSL target. Our model shows significant improvements over previous models like MERT and MusicFM, as demonstrated by its performance on the MARBLE benchmark. Additionally, experiments on the zero-shot music tagging task reveal that MuQ-MuLan achieves state-of-the-art music-text modal alignment. Our model code and checkpoints will be made open-source to support future research.

[1]https://github.com/LAION-AI/audio-dataset
[2]https://huggingface.co/datasets/sander-wood/m4-rag

VI. ETHIC DISCUSSION

We recognize the potential ethical concerns associated with MuQ. Throughout the data handling process, we've made it a priority to follow ethical guidelines, ensuring that personal information is protected and that the model remains free from biases. We also guarantee that no audio will be distributed or shared. We believe our work can positively impact the community by advancing automated music analysis, and MuQ can serve as a valuable tool for musicians and creators.

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